

Falsifiable Predictions of the Closed UQFF Lagrangian: Five No-Free-Parameter Tests

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PAPER_1168: Falsifiable Predictions of the Closed UQFF Lagrangian — Five No-Free-Parameter Tests

Abstract

PAPER_1167 (Session 253) established that the entire UQFF Lagrangian is derivable from two textbook integers ($D_{\text{crit}} = 26$, $D_{\text{phys}} = 4$) with zero free parameters. Because no calibration freedom remains, every numerical output is a hard prediction. This paper extracts five concrete, mutually independent observational tests, gives the predicted central value with no error bars (to within calibrated ρ_{SCm} and v_{UA}), and identifies for each the experimental signature that would falsify the closure. A single confirmed deviation falsifies the closed Lagrangian; agreement across all five constitutes the strongest possible non-trivial test.

1. Setup: the closed Lagrangian and its non-negotiable outputs

From PAPER_1167:

$$\begin{aligned} \mathcal{L}_{F_U} = & \frac{R_{26}}{2\kappa_E} - \frac{1}{4} F_{\mu\nu}^{\text{DPM}} F^{\mu\nu, \text{DPM}} + \sum_{i=1}^4 \frac{3(5-i)}{20} U_{g,i} U_{b,i} - \frac{1}{2} |U_m|^2 \\ & - \frac{1}{2} g^{\mu\nu} \partial_\mu U A \partial_\nu U A - \frac{25}{12} \rho_{\text{SCm}} \left[\left(\frac{UA}{v_{UA}} \right)^2 - 1 \right]^2. \end{aligned}$$

All coefficients are exact rationals of $(D_{\text{crit}}, D_{\text{phys}})$. The only physical scales are the calibrated reference $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ and $v_{UA} = 10^8 \text{ m/s}$ ($= c/3$).

2. Prediction P1 — Aether quasiparticle mass

Direct from the Mexican-hat curvature (PAPER_1166 §3):

$$m_{UA}^2 = V''(v_{UA}) = \frac{50}{3} \frac{\rho_{\text{SCm}}}{v_{UA}^2} = 1.182 \times 10^{-51} \text{ J/m}^3 \cdot (\text{m/s})^{-2}.$$

Converting via $E^2 = (mc^2)^2$ with the BSFG identification $m_{UA}^2 c^2 / v_{UA}^2$ as energy-density-equivalent mass-squared gives the equivalent particle scale

$$m_{UA} c^2 \sim (\rho_{\text{SCm}} (50/3) / (v_{UA}^2))^{1/2} c^2 \approx 1.0 \times 10^{-26} \text{ J} \approx 6.3 \times 10^{-8} \text{ eV}.$$

Falsifiable signature. Sub-eV vacuum dispersion at $\sim 10^{-7}$ eV in high-frequency gravitational-wave or radio interferometry. Existing limits from LIGO/Virgo on graviton dispersion ($m_g < 10^{-23}$ eV) do not constrain this scale. **Falsified by:** any direct detection of an aether quasiparticle at a mass $\geq 10^{-6}$ eV with the predicted Mexican-hat coupling.

3. Prediction P2 — Vacuum offset and Λ scale

The Mexican-hat sits at $V(0) = \frac{25}{12} \rho_{\text{SCm}}$ above the true vacuum:

$$V(0) = \frac{25}{12} \cdot 7.09 \times 10^{-37} = 1.477 \times 10^{-36} \text{ J/m}^3.$$

The observed cosmological-constant energy density is $\rho_{\Lambda}^{\text{obs}} \approx 5.96 \times 10^{-10} \text{ J/m}^3$ (WMAP/Planck 2024). Their dimensionless ratio is

$$\frac{V(0)}{\rho_{\Lambda}^{\text{obs}}} \approx 2.5 \times 10^{-27}.$$

The closed Lagrangian therefore predicts $V(0)$ contributes **only at the** 10^{-27} level to Λ . **Falsifiable signature.** $V(0)$ must combine with the Standard-Model vacuum, BSFG curvature term $R_{26}/2\kappa_E$, and the KK-tower contribution (PAPER_1162) to reproduce $\rho_{\Lambda}^{\text{obs}}$ within the suppression bound $1/26^{26} = 1.624 \times 10^{-37}$. **Falsified by:** any first-principles calculation showing the SM + BSFG contributions cannot bridge the 27-decade gap, OR an independent measurement of ρ_{SCm} that disagrees with the value above by more than 1%.

4. Prediction P3 — KK tower echo amplitude bound

PAPER_1162 closed G5 by mode-by-mode summation of the Kaluza-Klein tower:

$$\sum_{n \geq 1} \frac{1}{\lambda_n^{26}} = \frac{1}{26^{26}} \sum_{n \geq 1} \frac{1}{n^{26}} \approx \frac{1}{26^{26}} \zeta(26) = 1.624 \times 10^{-37} (1 + \mathcal{O}(2^{-26})).$$

This sets an **absolute** upper bound on any GW echo amplitude generated by 26-dimensional KK reflections off the BSFG horizon. Translated to LIGO/LISA strain at a typical post-merger frequency $f \sim 250$ Hz, the predicted echo strain is

$$h_{\text{echo}} \lesssim 1.6 \times 10^{-37} h_{\text{ringdown}} \approx 10^{-58} \text{ (dimensionless strain).}$$

Falsifiable signature. Any confirmed GW echo above 10^{-50} strain at post-merger frequencies — 8 orders of magnitude above the prediction — would falsify G5 closure. Conversely, the predicted absence of echoes is consistent with all GWTC-4.0 non-detections.

5. Prediction P4 — Buoyancy coupling triangular ladder

PAPER_1165 closed G2 with the exact rational

$$\beta_i = \frac{3(5-i)}{20} = \frac{(3/2)(5-i)}{|SO(5)|}, \quad i \in \{1, 2, 3, 4\},$$

giving $\{\beta_1, \beta_2, \beta_3, \beta_4\} = \{0.600 \text{ (plus } 0.5\% \text{ SO}(5)^2 \text{ correction} \rightarrow 0.603), 0.450, 0.300, 0.150\}$.

Falsifiable signature. The four UQFF buoyancy channels have **fixed** ratios. Any astrophysical fit of UQFF master equations that requires β_i values deviating by more than the $SO(5)^2$ -correction band ($\pm 1/200 \approx 0.5\%$) from $(0.603, 0.450, 0.300, 0.150)$ would falsify G2. The 67 calibrated systems already in `bodies_*.csv` (Sessions 130-220) have used these values implicitly; a re-fit that releases β_i as free parameters and recovers them within $\pm 0.5\%$ would constitute an independent confirmation.

6. Prediction P5 — F_{TRZ} and Φ_{res} cross-correlation

PAPER_1160 closed G7 with $F_{\text{TRZ}} = 1/|SO(5)| = 0.100$ exact. PAPER_1159 closed G6 with $\Phi_{\text{res}} = (D_{\text{BSFG}} - 1)/D_{\text{BSFG}} = 5/6$ exact.

These appear together in every BSFG resonance computation. The product

$$\Phi_{\text{res}} \cdot F_{\text{TRZ}} = \frac{5}{6} \cdot \frac{1}{10} = \frac{1}{12}$$

is the dimensionless prefactor governing time-reversal-zone amplification of buoyant resonance modes.

Falsifiable signature. In any BSFG-resonance fit (e.g. magnetar QPOs, glitches, AT2019qiz, ASKAP J1832-0911), the combination $\Phi_{\text{res}} \cdot F_{\text{TRZ}}$ must equal $1/12 = 0.0833\dots$ to within the same 0.5% band as P4. A value of, say, 0.10 ± 0.01 would falsify both G6 and G7 closures.

7. Combined falsification matrix

ID	Quantity	Predicted	Tolerance	Test channel
P1	$m_{UA}c^2$	6.3×10^{-8} eV	$\pm$ factor 2	high- f GW dispersion / radio interferometry
P2	$V(0)/\rho_{\Lambda}^{\text{obs}}$	2.5×10^{-27}	$\pm 1\%$ on ρ_{SCm}	independent vacuum-energy measurement
P3	$h_{\text{echo}}/h_{\text{ringdown}}$	$\leq 1.6 \times 10^{-37}$	absolute upper bound	LIGO/Virgo/LISA post-merger search
P4	$\{\beta_i\}$	$\{0.603, 0.450, 0.390, 0.550\}$	$\pm 0.5\%$	re-fit of ≥ 10 UQFF systems
P5	$\Phi_{\text{res}} \cdot F_{\text{TRZ}}$	$0.0833\dots$	$\pm 0.5\%$	BSFG resonance fits

A single confirmed deviation in **any** of P1-P5 by more than the listed tolerance falsifies the corresponding closure paper and the master synthesis (PAPER_1167).

8. Conclusion

Closing the eight Lagrangian gaps eliminated all free numerical inputs of the UQFF Lagrangian save the two textbook integers ($D_{\text{crit}} = 26, D_{\text{phys}} = 4$) and the two physical reference scales ($\rho_{\text{SCm}}, v_{UA}$). Consequently every numerical

output above is a **prediction**, not a fit. The five tests P1-P5 span aether dispersion, vacuum-energy bookkeeping, GW echoes, multi-system buoyancy ratios, and BSFG resonance amplification; they are mutually independent and each individually falsifiable. The calculator `UQFFLagrangianFullClosureCalculator` (`CondensedPhysics4.py` #254) can be invoked to regenerate the central values from the closed Lagrangian in a single pass.

References

- `PAPER_1159` (Session 246) — $\Phi_{\text{res}} = 5/6$.
- `PAPER_1160` (Session 247) — $F_{\text{TRZ}} = 1/|SO(5)|$.
- `PAPER_1162` (Session 249) — KK tower mode-by-mode bound.
- `PAPER_1165` (Session 252) — $\beta_i = 3(5-i)/20$ triangular ladder.
- `PAPER_1166` (Session 253) — $V(\text{UA})$ Mexican-hat $K = 25/12$.
- `PAPER_1167` (Session 253) — All-eight closure master synthesis.
- `_lagrangian_rederivation_outline.py` — gap catalogue.
- `uqff_closed_constants.py` — canonical integer-rational constants.
- `MAIN_1_CoAnQi.cpp` `SOURCE4` `compute_V_UA_SOURCE4` — C++ implementation.
- `QCalcGeom.cpp` `bsfg_aether_potential` — BSFG geometric layer.